

Chapter 9: The number plane

6 lessons · one lesson for each exercise · each lesson is
one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-LIN-C-01 creates and
displays number patterns and graphs linear relationships on the
Cartesian plane.

CHAPTER

9

9.01 Grid references on maps

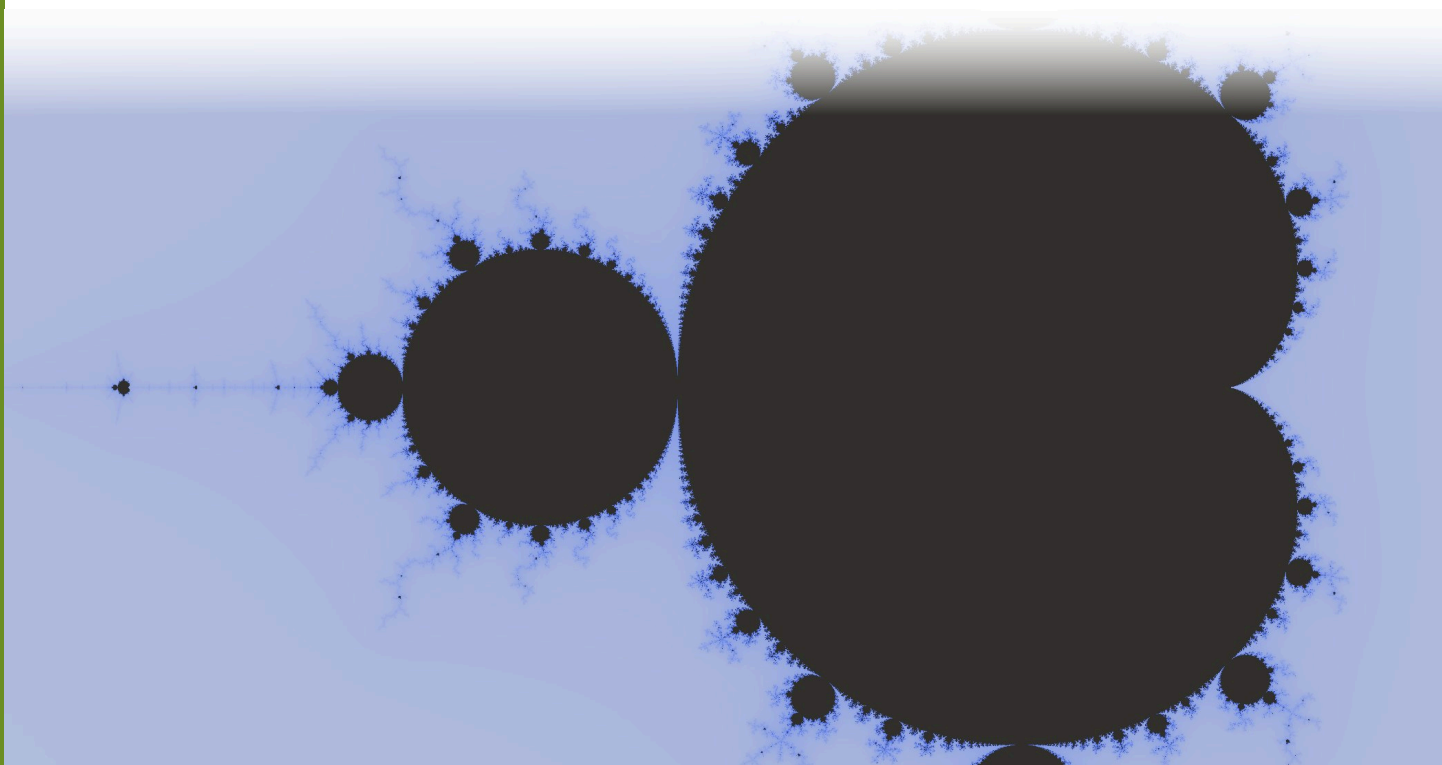
9.02 Points in the first quadrant

9.03 The number plane

9.04 Plotting tables of values

9.05 Tables of values from rules

9.06 Graphing linear equations



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

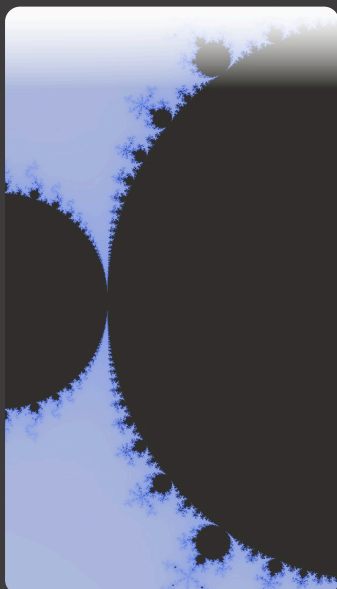
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
9.01	Grid references on maps	3		☐
9.02	Points in the first quadrant	9		☐
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9.04	Plotting tables of values	21		☐
9.05	Tables of values from rules	27		☐
9.06	Graphing linear equations	33		☐

In this chapter you will

- use grid references on a map
- plot and read points in the first quadrant
- plot and read points in all four quadrants
- plot points from tables and describe patterns
- complete tables of values from rules
- graph linear equations and test whether a point is on a line



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the whole set, drawn on the complex number plane. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
9.01

Grid references on maps

Date _____

Learning intentions

We are learning to:

- use grid references to find and describe locations on a map

Success criteria

I can:

- ☐ read a grid reference: letter (across) then number (up)
- ☐ state the feature at a grid reference
- ☐ write the grid reference of a feature

Key terms Write the definition as your teacher explains it.

Grid reference

Map

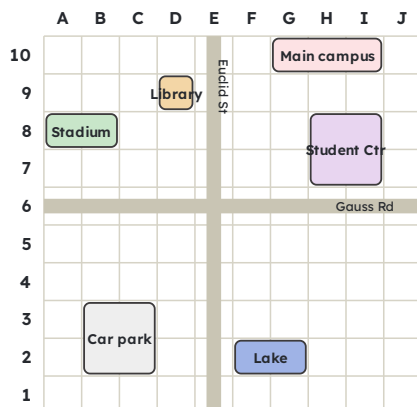
Notes

Level 1 · Getting started

Easy · 30 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the map. What is at each grid reference?



a What is at F2?

Your teacher works this out. Copy every step.

b What is at D9?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **30 QUESTIONS**

REMEMBER A grid reference names a square.

ROUND 1 Questions 1–6: write the grid reference: the column letter first, then the row number.

1 column A, row 5

2 column D, row 1

3 column E, row 4

4 column G, row 5

5 column F, row 6

6 column H, row 6

YOU DO On your own: **Easy** keep going: the next rounds.

24 QUESTIONS MORE

ROUND 2 Questions 7–18: use the map. What is at each grid reference?

6	Wharf	Fire		Post		Vet		Hotel
5					Club			Beach
4	Farm		Mall				Oval	
3				Gym			Cafe	
2		Shop			Bus			Zoo
1	Pool		Bank			Park		
	A	B	C	D	E	F	G	H

7 H2

8 D3

9 C1

10 G3

11 F1

12 E2

13 D6

14 G4

15 A6

16 A4

17 A1

18 B6

ROUND 3 Questions 19–30: write the grid reference of each place.

6	Wharf	Fire		Post		Vet		Hotel
5					Club			Beach
4	Farm		Mall				Oval	
3				Gym			Cafe	
2		Shop			Bus			Zoo
1	Pool		Bank			Park		
	A	B	C	D	E	F	G	H

19 Cafe

20 Club

21 Shop

22 Mall

23 Zoo

24 Hotel

25 Pool

26 Bus

27 Fire

28 Vet

29 Beach

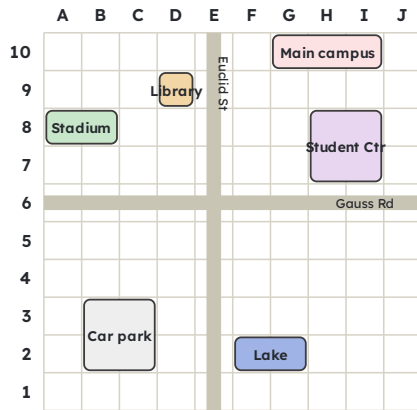
30 Park

Level 2 · Building up

Medium · 10 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the map. Write the grid references.



a Write the grid reference of the library.

Your teacher works this out. Copy every step.

b Which grid squares does the car park cover?

Your teacher works this out. Copy every step.

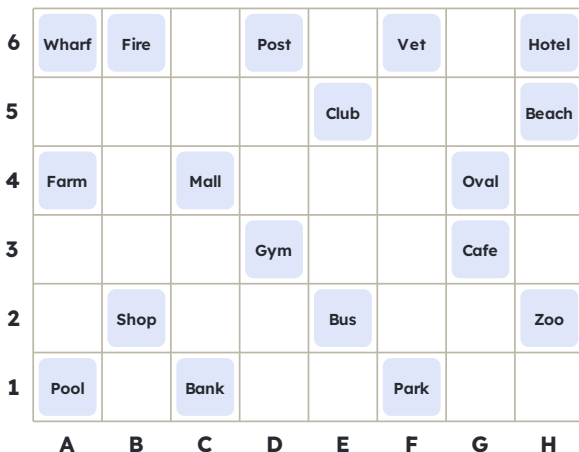
YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

10 QUESTIONS

REMEMBER Letter first (across), then number (up).

ROUND 1 Questions 1–10: start at the place given and move. Where do you end up?



1 Bank: 1 left, 4 up

2 Park: 1 left, 5 up

3 Bus: 3 right, 1 up

4 Post: 1 right, 1 down

5 Pool: 5 right, 4 up

6 Zoo: 1 left, 2 up

7 Oval: 1 left, 1 down

8 Gym: 1 right, 2 up

9 Wharf: 2 right, 1 down

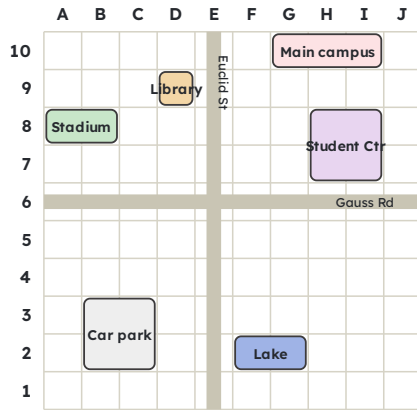
10 Farm: 6 right, 1 down

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use the map, then explain.



a Describe how to walk from the lake to the library along the roads.

Your teacher works this out. Copy every step.

b Why is the letter written before the number?

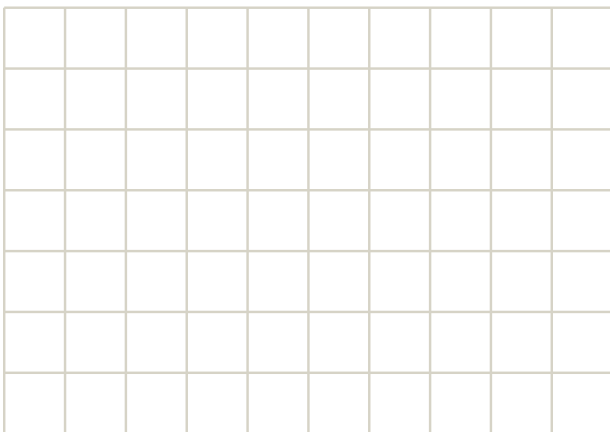
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Draw your own.

2 QUESTIONS

REMEMBER Large features may cover several squares.

1 Draw a map of your school on the grid. Mark 5 places and list their grid references.



2 Write directions from one place to another using grid references.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A treasure is 3 squares right and 4 squares up from B2. Where is it?

1. 3 squares right of B is E
2. 4 squares up from 2 is 6

Answer: E6

YOU DO

On your own: read the worked example, then try these.

- E1** From H7, go 5 left and 2 down. Where are you?

- E2** How many squares is it from A1 to J10 if you can only move up or right?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is at A8?

Level 2

Write the grid reference of the lake.

Level 3

From D9 go 4 right and 2 down. What is there?

Lesson summary

Missed the lesson? Start here.

Remember

1. A grid reference names a square.
2. Letter first (across), then number (up).
3. Large features may cover several squares.
4. Check by running your finger across, then up.

Level 1

What is at F2?

1. Column F, row 2

Answer: the lake

Level 2

Library

1. Column D, row 9

Answer: D9

Level 3

Car park

1. Covers 4 squares

Answer: B2, C2, B3, C3

Extension

3 right, 4 up from B2

1. B → E, 2 → 6

Answer: E6

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
9.02

Points in the first quadrant

Date _____

Learning intentions

We are learning to:

- plot and read points on a number plane in the first quadrant

Success criteria

I can:

- ☐ know the **x-coordinate** is across and the **y-coordinate** is up
- ☐ plot a point (x, y)
- ☐ read the coordinates of a point

Key terms Write the definition as your teacher explains it.

Coordinates

Origin

Axis

Notes

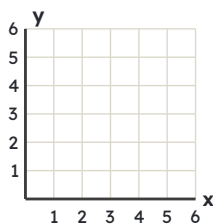
Level 1 · Getting started

Easy · 24 questions

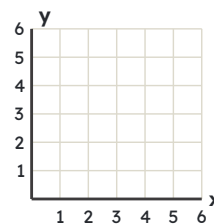
WE DO With your teacher: copy what they write on the board.

Example 1 Plot and label the points.

a Plot A(2, 3) and B(1, 6).



b Plot C(3, 0) and D(0, 4).



YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER (x, y): x across first, then y up.

ROUND 1 Questions 1–12: write the x-coordinate (x) or the y-coordinate (y) of the point.

1 y of (2, 6)

2 y of (2, 10)

3 y of (12, 4)

4 x of (1, 9)

5 y of (7, 12)

6 x of (12, 0)

7 x of (6, 4)

8 x of (7, 9)

9 x of (8, 0)

10 x of (4, 1)

11 y of (9, 10)

12 x of (0, 10)

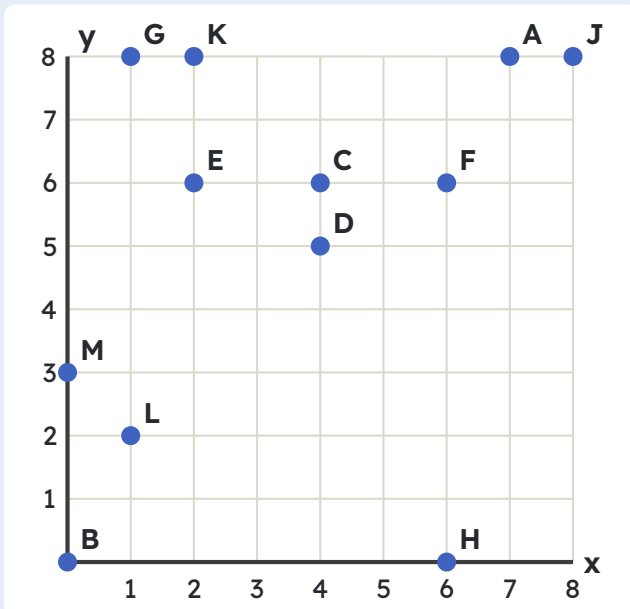
YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 13–24: write the coordinates of each point.



13 A

14 B

15 C

16 D

17 E

18 F

19 G

20 H

21 J

22 K

23 L

24 M

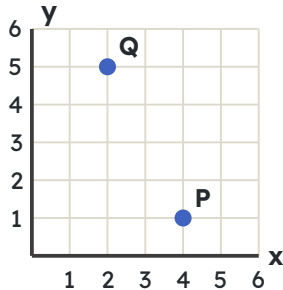
Level 2 · Building up

Medium · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Read the coordinates, then explain.

a Write the coordinates of P and Q.



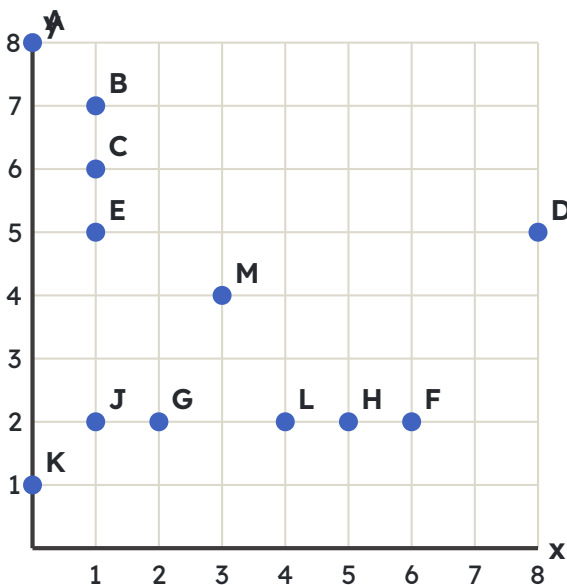
b Is (2, 5) the same point as (5, 2)? Explain.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **12 QUESTIONS**

REMEMBER Start at the origin (0, 0).

ROUND 1 Questions 1–12: which point is at these coordinates?



① (1, 7)

② (1, 2)

③ (3, 4)

④ (6, 2)

⑤ (1, 6)

⑥ (1, 5)

⑦ (8, 5)

⑧ (5, 2)

⑨ (0, 1)

⑩ (4, 2)

⑪ (0, 8)

⑫ (2, 2)

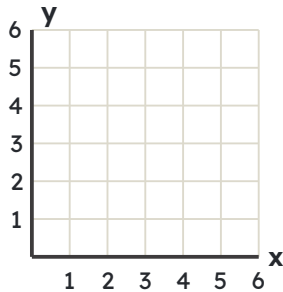
Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Plot, join and name the shape.

- a** Plot $(1, 1)$, $(5, 1)$, $(5, 4)$, $(1, 4)$ and join them. Name the shape.



- b** What are the coordinates of the origin?

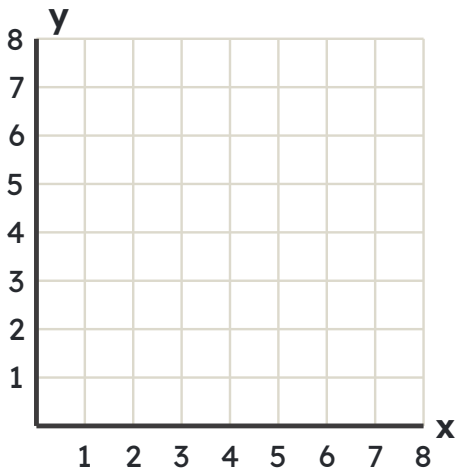
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

2 QUESTIONS

REMEMBER Points on the x-axis have $y = 0$.

- 1** Three corners of a square are $(1, 2)$, $(5, 2)$ and $(5, 6)$. Plot them and find the fourth corner.



- 2** Name 3 points on the x-axis and 3 on the y-axis. What do you notice?



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Find the midpoint of (2, 3) and (6, 7).

1. Halfway across: $(2 + 6) \div 2 = 4$
2. Halfway up: $(3 + 7) \div 2 = 5$

Answer: (4, 5)

YOU DO

On your own: read the worked example, then try these.

- E1 Find the midpoint of (1, 8) and (5, 2).

- E2 A rectangle has corners (1, 1) and (7, 5). Find its area in square units.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is the x-coordinate of (7, 3)?

Level 2

Where is the point (0, 5)?

Level 3

A rectangle has corners (1, 2), (6, 2), (6, 5). Find the fourth corner.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. (x, y) : x across first, then y up.
 3. Points on the x -axis have $y = 0$.

2. Start at the origin $(0, 0)$.
4. Points on the y -axis have $x = 0$.

Level 1

Plot $A(2, 3)$

1. 2 across, 3 up

Answer: a dot labelled A

Level 2

Coordinates of P

1. 4 across, 1 up

Answer: $(4, 1)$

Level 3

$(2, 5)$ vs $(5, 2)$

1. Order matters

Answer: different points

Extension

Midpoint of $(2, 3)$, $(6, 7)$

1. Average x and y

Answer: $(4, 5)$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.03

The number plane

Date _____

Learning intentions

We are learning to:

- plot and read points in all four quadrants

Success criteria

I can:

- ☐ name the four quadrants
- ☐ plot points with negative coordinates
- ☐ state which quadrant a point is in from its signs

Key terms Write the definition as your teacher explains it.

Quadrant

Origin

Notes

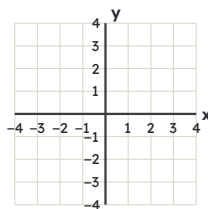
Level 1 · Getting started

Easy · 24 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Label the quadrants, then name the quadrant.

a Label the quadrants 1st to 4th.



b In which quadrant is $(-3, -3)$?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **24 QUESTIONS**

REMEMBER The axes cross at the origin $(0, 0)$.

ROUND 1 Questions 1–12: which quadrant (1st, 2nd, 3rd or 4th) or which axis is the point on?

1 $(-2, -9)$

2 $(-2, 8)$

3 $(-6, 9)$

4 $(4, -8)$

5 $(6, -5)$

6 $(-6, -4)$

7 $(-7, -6)$

8 $(-1, 7)$

9 $(-8, 4)$

10 $(1, -5)$

11 $(-4, 7)$

12 $(-2, -8)$

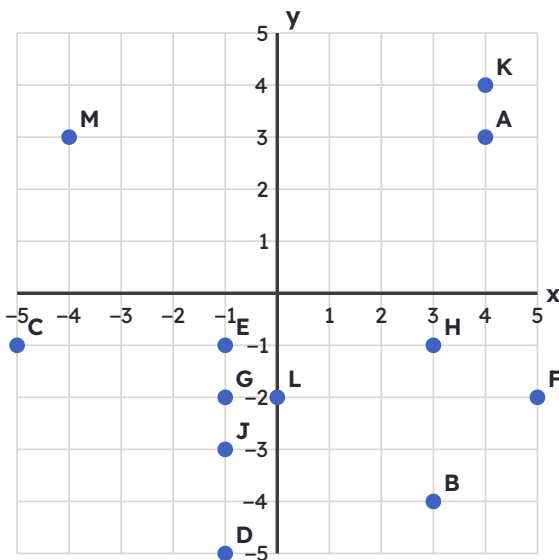
YOU DO

On your own: **Easy** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2

Questions 13–24: write the coordinates of each point.



13 A

14 B

15 C

16 D

17 E

18 F

19 G

20 H

21 J

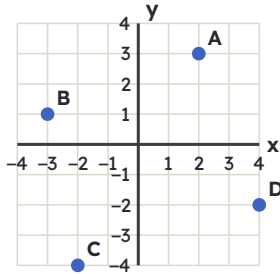
22 K

23 L

24 M

Level 2 • Building up

Medium • 28 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Read the coordinates, then name the quadrant.**a** Write the coordinates of each point.**b** Which quadrant has a positive x-coordinate and a negative y-coordinate?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line. **28 QUESTIONS****REMEMBER** 1st (+, +), 2nd (-, +), 3rd (-, -), 4th (+, -).**ROUND 1** Questions 1–16: reflect the point in the axis given.**1** $(-7, -7)$ in the y-axis**2** $(-3, 3)$ in the y-axis**3** $(4, 2)$ in the y-axis**4** $(-4, -1)$ in the x-axis**5** $(1, 4)$ in the y-axis**6** $(4, -6)$ in the x-axis**7** $(7, -3)$ in the y-axis**8** $(4, 4)$ in the y-axis**9** $(-4, -8)$ in the y-axis**10** $(-2, -6)$ in the y-axis**11** $(8, -7)$ in the x-axis**12** $(5, -6)$ in the x-axis**13** $(6, 8)$ in the x-axis**14** $(-2, 4)$ in the x-axis**15** $(-1, -8)$ in the x-axis**16** $(5, 2)$ in the y-axis

YOU DO On your own: **Medium** keep going: Round 2.

12 QUESTIONS MORE

ROUND 2 Questions 17–28: which quadrant is described?

- | | | |
|--|---|---|
| 17 both positive
_____ | 18 $x = 0$
_____ | 19 right of the y-axis, above the x-axis
_____ |
| 20 left of the y-axis, above the x-axis
_____ | 21 right of the y-axis, below the x-axis
_____ | 22 x positive, y negative
_____ |
| 23 x negative, y positive
_____ | 24 $y = 0$
_____ | 25 both negative
_____ |
| 26 $(-a, a)$ for a positive number a
_____ | 27 left of the y-axis, below the x-axis
_____ | 28 x and y are both zero
_____ |

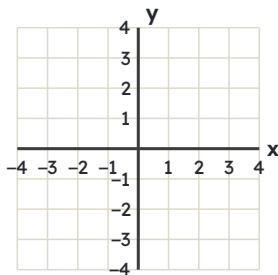
Level 3 · Challenge

Challenging · 3 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Plot and join, then write points.

- a** Plot $(1, 3)$, $(2, 2)$, $(3, 1)$, $(2, 0)$, $(1, -1)$, $(0, -2)$, $(-1, -1)$ and join.
- b** Write a point in the 2nd quadrant and a point on the y-axis.



Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Think it through.

3 QUESTIONS

REMEMBER Quadrants go anticlockwise from top right.

- 1 In which quadrant is a point with a negative x-coordinate and a positive y-coordinate? A 1st B 2nd C 3rd D 4th
- 2 What are the coordinates of the origin?
- 3 Reflect $(3, -2)$ in the y-axis. Write the new point.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

A square has corners $(-2, -1)$ and $(3, -1)$ along its bottom. Find the other two corners (above).

1. Side length = $3 - (-2) = 5$
2. Go up 5: $y = -1 + 5 = 4$

Answer: $(3, 4)$ and $(-2, 4)$ **YOU DO**

On your own: read the worked example, then try these.

- E1** A triangle has corners $(-3, 0)$, $(3, 0)$ and $(0, 4)$. Find its area.

- E2** Reflect the triangle $(1, 1)$, $(4, 1)$, $(1, 3)$ in the x-axis. Write the new corners.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO**Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

In which quadrant is $(-4, 2)$?

Level 2

Write the coordinates of a point in the 4th quadrant.

Level 3

Reflect $(-2, 5)$ in the x-axis.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. The axes cross at the origin (0, 0).
 2. 1st (+, +), 2nd (–, +), 3rd (–, –), 4th (+, –).
 3. Quadrants go anticlockwise from top right.
 4. Points on an axis are in no quadrant.

Level 1

(–3, –3)

1. Both negative

Answer: 3rd quadrant

Level 2

Point B

1. 3 left, 1 up

Answer: (–3, 1)

Level 3

+x, –y

1. Bottom right

Answer: 4th quadrant

Extension

Square on (–2, –1), (3, –1)

1. Side 5, go up

Answer: (3, 4), (–2, 4)

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.04

Plotting tables of values

Date _____

Learning intentions

We are learning to:

- plot points from a table and describe the pattern

Success criteria

I can:

- ☐ read (x, y) pairs from a table
- ☐ plot the points and join them with a ruler
- ☐ describe the pattern (straight line, vertical, horizontal, increasing, decreasing)

Key terms Write the definition as your teacher explains it.

Table of values

Pattern

Notes

Level 1 · Getting started

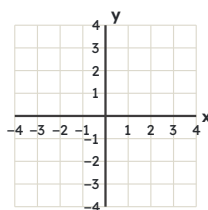
Easy · 19 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Plot the table, then describe the pattern.

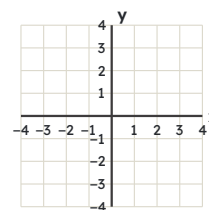
a Plot the table and describe the pattern.

x	-2	-2	-2	-2
y	-2	-1	0	1



b Plot and describe.

x	-2	-1	0	2
y	3	3	3	3



YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

19 QUESTIONS

REMEMBER Each column of a table is a point (x, y).

ROUND 1 Questions 1–4: write each column of the table as a point (x, y).

1 x: 0, 1, 2, 3
y: 5, 9, 13, 17

2 x: 0, 1, 2, 3
y: 0, 3, 6, 9

3 x: 0, 1, 2, 3
y: 5, 7, 9, 11

4 x: 0, 1, 2, 3
y: 2, 3, 4, 5

YOU DO

On your own: **Easy** keep going: Round 2.

15 QUESTIONS MORE

ROUND 2

Questions 5–19: the points follow a pattern. Write the next point.

5 (3, 13), (4, 17), (5, 21)

6 (1, 7), (2, 10), (3, 13)

7 (1, 1), (2, 2), (3, 3)

8 (3, 18), (4, 22), (5, 26)

9 (0, 4), (1, 8), (2, 12)

10 (0, 2), (1, 3), (2, 4)

11 (0, 6), (1, 9), (2, 12)

12 (1, 5), (2, 9), (3, 13)

13 (3, 9), (4, 11), (5, 13)

14 (1, 3), (2, 6), (3, 9)

15 (0, 5), (1, 8), (2, 11)

16 (3, 6), (4, 7), (5, 8)

17 (0, 4), (1, 6), (2, 8)

18 (1, 6), (2, 9), (3, 12)

19 (3, 16), (4, 20), (5, 24)

Level 2 · Building up

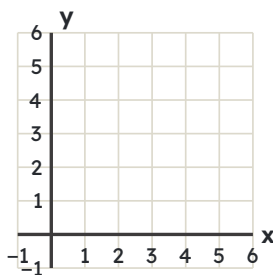
Medium · 15 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Plot the table, then describe the pattern.

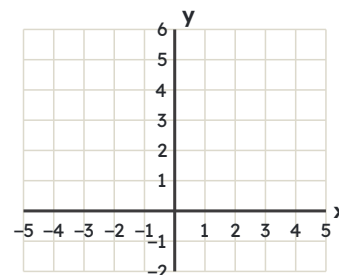
a Plot and describe.

x	0	1	2	5
y	5	4	3	0



b Plot and describe.

x	-1	0	1	2
y	-1	1	3	5



YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

REMEMBER Plot each point carefully.

ROUND 1 Questions 1–15: the points follow a pattern. Find the missing coordinate.

1 (0, -1), (1, 4), (2, 9), (5, ?)

2 (0, -2), (1, 1), (2, 4), (5, ?)

3 (0, 0), (1, 3), (2, 6), (5, ?)

4 (0, 4), (1, 7), (2, 10), (5, ?)

5 (0, 4), (1, 9), (2, 14), (5, ?)

6 (0, 3), (1, 8), (2, 13), (5, ?)

7 (0, -2), (1, -1), (2, 0), (5, ?)

8 (0, 4), (1, 8), (2, 12), (5, ?)

9 (0, 0), (1, 5), (2, 10), (5, ?)

10 (0, 0), (1, 2), (2, 4), (5, ?)

11 (0, -1), (1, 1), (2, 3), (5, ?)

12 (0, -1), (1, 3), (2, 7), (5, ?)

13 (0, 3), (1, 4), (2, 5), (5, ?)

14 (0, 3), (1, 5), (2, 7), (5, ?)

15 (0, -2), (1, 0), (2, 2), (5, ?)

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the pattern, then predict.

a In the table in Example 4, what happens to y when x goes up by 1?

Your teacher works this out. Copy every step.

b Predict y when $x = 3$ in Example 4.

Your teacher works this out. Copy every step.

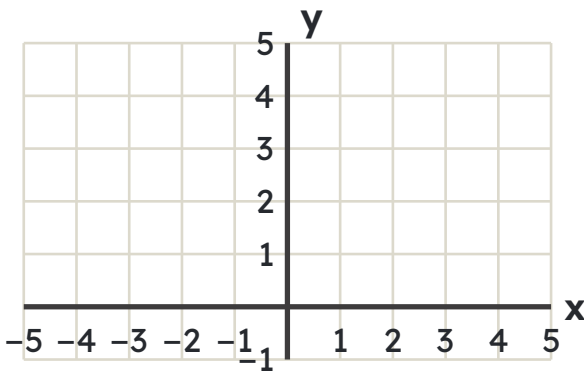
YOU DO On your own: **Challenging** Plot and describe.

2 QUESTIONS

REMEMBER Join them with a ruler if they line up.

1

x	-2	-1	0	1	2
y	4	1	0	1	4



2 Is the pattern in Question 1 a straight line? Describe it.

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

The points (0, 2), (1, 5), (2, 8) follow a pattern.
Find y when $x = 10$.

1. y goes up by 3 each time x goes up by 1
2. Start at 2 when $x = 0$: $y = 3x + 2$
3. $3 \times 10 + 2$

Answer: 32

YOU DO

On your own: read the worked example, then try these.

- E1 The points (0, 7), (1, 5), (2, 3) follow a pattern.
Find y when $x = 6$.

- E2 Find the rule for the points (1, 4), (2, 8), (3, 12).

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Write the points in this table: $x = 0, 1, 2$ and $y = 3, 5, 7$.

Level 2

Describe the pattern of the points (1, 4), (2, 4), (3, 4).

Level 3

What happens to y each time x goes up by 1 for (0, 1), (1, 4), (2, 7)?

Lesson summary

Missed the lesson? Start here.

Remember

1. Each column of a table is a point (x, y).
2. Plot each point carefully.
3. Join them with a ruler if they line up.
4. Describe: horizontal, vertical, going up or going down.

Level 1

x always -2

1. All points above/below each other

Answer: vertical line

Level 2

y always 3

1. All points level

Answer: horizontal line

Level 3

(0, 5), (1, 4), (2, 3)

1. y goes down 1

Answer: line going down

Extension

(0, 2), (1, 5), (2, 8)

1. $y = 3x + 2$

Answer: 32 when $x = 10$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.05

Tables of values from rules

Date _____

Learning intentions

We are learning to:

- complete a table of values from a rule

Success criteria

I can:

- ☐ substitute each x value into the rule
- ☐ use negative numbers carefully
- ☐ write the rule from a table

Key terms Write the definition as your teacher explains it.

Rule

Substitute

Notes

Level 1 · Getting started

Easy · 19 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Complete each table from its rule.

a Complete for $y = x + 3$.

x	-1	0	1	2	3
y					

Your teacher works this out. Copy every step.

b Complete for $q = 2p - 7$.

p	-2	-1	0	1	2
q					

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line. **19 QUESTIONS**

REMEMBER Write the rule.

ROUND 1 Questions 1–9: substitute to find y .

1 $y = 2x + 1, x = 7$

2 $y = 4x + 6, x = 2$

3 $y = 4x + 8, x = 2$

4 $y = 5x + 1, x = 1$

5 $y = 6x - 2, x = 4$

6 $y = x - 2, x = 4$

7 $y = 5x, x = 1$

8 $y = 3x - 4, x = 3$

9 $y = 4x + 3, x = 4$

YOU DO

On your own: **Easy** keep going: Round 2.

10 QUESTIONS MORE

ROUND 2

Questions 10–19: complete the table: write the y-values for $x = 0, 1, 2, 3$.

10 $y = 3x + 8$

11 $y = 4x + 4$

12 $y = x + 8$

13 $y = 3x + 9$

14 $y = 4x + 3$

15 $y = x + 6$

16 $y = 4x + 6$

17 $y = 5x$

18 $y = 2x + 3$

19 $y = x + 9$

Level 2 · Building up

Medium · 15 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Complete each table from its rule.

a Complete for $d = 5 - c$.

c	-1	0	1	2	3
d					

Your teacher works this out. Copy every step.

b Complete for $y = 3x + 1$.

x	-2	-1	0	1	2
y					

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

15 QUESTIONS

REMEMBER Replace x with each value (use brackets for negatives).

ROUND 1 Questions 1–15: find x for the value of y given.

1 $y = 5x + 5, y = 15$

2 $y = 5x + 3, y = 48$

3 $y = 3x + 5, y = 29$

4 $y = 4x + 7, y = 11$

5 $y = 4x + 8, y = 28$

6 $y = 4x + 1, y = 33$

7 $y = 2x + 2, y = 10$

8 $y = 3x + 4, y = 7$

9 $y = 6x + 8, y = 8$

10 $y = 6x + 3, y = 39$

11 $y = 5x, y = 30$

12 $y = 3x + 7, y = 28$

13 $y = 3x + 4, y = 34$

14 $y = 4x + 7, y = 7$

15 $y = 4x + 2, y = 38$

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the rule for each table.

a Find the rule.

x	0	1	2	3
y	4	6	8	10

Your teacher works this out. Copy every step.

b Find the rule.

x	1	2	3	4
y	7	6	5	4

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Find the rule for each table.

2 QUESTIONS

REMEMBER Calculate y.

1

x	0	1	2	3
y	4	6	8	10

2

x	1	2	3	4
y	7	6	5	4

Extension · Stretch yourself

Finished the challenge? Stretch yourself here.

Worked example

Complete a table for $y = x^2 - 1$ using $x = -2$ to 2.

1. Square first, then subtract 1
2. $(-2)^2 - 1 = 3$

Answer: $y = 3, 0, -1, 0, 3$

YOU DO

On your own: read the worked example, then try these.

- E1 Complete a table for $y = x^2 + x$ using $x = -2$ to 2.

- E2 Find the rule: $x = 1, 2, 3, 4$ and $y = 1, 4, 9, 16$.

YOU DO Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

For $y = x + 5$, find y when $x = -2$.

Level 2

For $y = 4x - 1$, complete for $x = 0, 1, 2$.

Level 3

Find the rule: $x = 0, 1, 2$ and $y = 1, 4, 7$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Write the rule.
2. Replace x with each value (use brackets for negatives).
3. Calculate y .
4. To find a rule: how much does y change when x goes up 1? Then find the start value.

Level 1

$$y = x + 3$$

1. $-1 + 3 = 2, \dots$

Answer: 2, 3, 4, 5, 6

Level 2

$$q = 2p - 7$$

1. $2 \times (-2) - 7 = -11$

Answer: $-11, -9, -7, -5, -3$

Level 3

Rule for 4, 6, 8, 10

1. $+2$ each time, start 4

Answer: $y = 2x + 4$

Extension

$$y = x^2 - 1$$

1. Square first

Answer: 3, 0, -1 , 0, 3

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
9.06

Graphing linear equations

Date _____

Learning intentions

We are learning to:

- graph a linear equation using a table of values
- test whether a point lies on a line

Success criteria

I can:

- ☐ complete a table of values for the equation
- ☐ plot the points and rule a straight line through them
- ☐ substitute a point into the equation to test it

Key terms Write the definition as your teacher explains it.

Linear

Equation

Notes

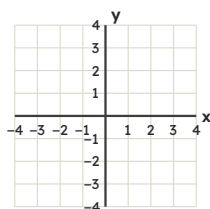
Level 1 · Getting started

Easy · 17 questions

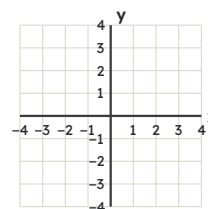
WE DO With your teacher: copy what they write on the board.

Example 1 Complete a table, then graph the line.

a Graph $y = 2x - 1$ for $x = -1$ to 2.



b Graph $y = -x + 3$.



YOU DO On your own: **Easy** practise the skill. Work down the page, one round at a time. Write each answer on its line.

17 QUESTIONS

REMEMBER Make a table with 3 or more x values.

ROUND 1 Questions 1–9: does the point lie on the line? Write yes or no.

1 (0, 2) on $y = 4x + 2$

2 (2, 7) on $y = 2x + 3$

3 (3, 13) on $y = 4x + 1$

4 (–3, 2) on $y = x + 3$

5 (3, 14) on $y = 3x + 5$

6 (0, 1) on $y = x + 1$

7 (–1, 1) on $y = 3x + 4$

8 (3, 8) on $y = x + 4$

9 (0, –1) on $y = 4x - 1$

YOU DO

On your own: **Easy** keep going: Round 2.

8 QUESTIONS MORE

ROUND 2

Questions 10–17: write the points to plot for $x = -1, 0, 1, 2$.

10 $y = 2x - 1$

11 $y = 3x - 1$

12 $y = x + 3$

13 $y = 2x + 2$

14 $y = 3x + 1$

15 $y = x$

16 $y = 3x$

17 $y = 3x + 2$

Level 2 · Building up

Medium · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Does the point lie on the line?

a Does (1, 5) lie on $y = 3x + 2$?

Your teacher works this out. Copy every step.

b Does (-2, -4) lie on $y = 3x + 2$?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** practise the skill. Work down the page, one round at a time. Write each answer on its line.

16 QUESTIONS

REMEMBER Substitute to find y .

ROUND 1 Questions 1–16: the point lies on the line. Find the missing coordinate.

1 (0, ?) on $y = -2x - 4$

2 (-1, ?) on $y = x + 4$

3 (0, ?) on $y = 2x$

4 (2, ?) on $y = 2x$

5 (4, ?) on $y = -x - 2$

6 (-2, ?) on $y = 2x + 4$

7 (3, ?) on $y = 4x - 1$

8 (0, ?) on $y = 2x + 2$

9 (-1, ?) on $y = 4x$

10 (-3, ?) on $y = 2x - 4$

11 (4, ?) on $y = -x + 3$

12 (2, ?) on $y = 3x + 6$

13 (3, ?) on $y = 4x - 3$

14 (4, ?) on $y = 4x + 3$

15 (0, ?) on $y = 3x - 1$

16 (-4, ?) on $y = -3x - 2$

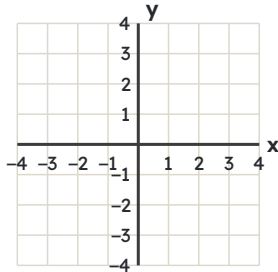
Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Graph the line, then answer.

a Graph $y = \frac{1}{2}x - 2$ for $x = -4$ to 4.



b Where does $y = 2x - 1$ cross the y-axis?

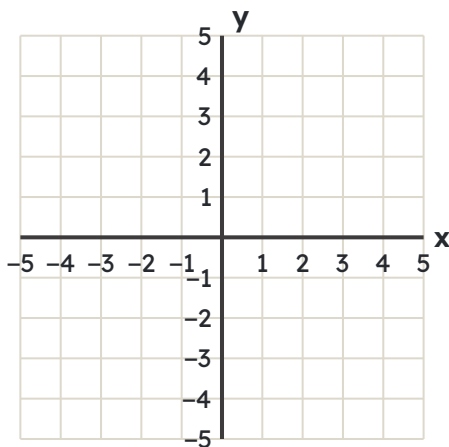
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Graph and compare.

2 QUESTIONS

REMEMBER Plot and rule a line through all the points.

1 Graph $y = x$ and $y = x + 2$ on the same plane. What do you notice?



2 Where do $y = x + 1$ and $y = -x + 3$ meet? (Use your graphs.)



Extension · Stretch yourself

Finished the challenge? Stretch yourself here.



Worked example

Find the value of k if $(2, k)$ lies on $y = 4x - 3$.

1. Substitute $x = 2$
2. $k = 4 \times 2 - 3$

Answer: $k = 5$

YOU DO

On your own: read the worked example, then try these.

- E1 Find m if $(m, 11)$ lies on $y = 2x + 5$.

- E2 Which of these lines are parallel: $y = 2x + 1$, $y = x + 2$, $y = 2x - 3$?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

For $y = x + 4$, find y when $x = 1$.

Level 2

Does $(2, 5)$ lie on $y = 2x + 1$?

Level 3

Does $(-1, 3)$ lie on $y = -2x + 1$?

Lesson summary

Missed the lesson? Start here.

Remember

1. Make a table with 3 or more x values.
2. Substitute to find y .
3. Plot and rule a line through all the points.
4. Test a point: substitute x and see if you get its y .

Level 1

$$y = 2x - 1$$

1. x : -1, 0, 1, 2 $\rightarrow$ y : -3, -1, 1, 3

Answer: a straight line through (0, -1)

Level 2

$$(1, 5) \text{ on } y = 3x + 2?$$

1. $3 \times 1 + 2 = 5$ ✓

Answer: yes

Level 3

$$(-2, -4)$$

1. $3 \times (-2) + 2 = -4$ ✓

Answer: yes

Extension

$$(2, k) \text{ on } y = 4x - 3$$

1. $8 - 3$

Answer: $k = 5$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.